

Task-03: Bank Marketing Prediction using Decision Tree

Abstract

This project predicts whether a bank customer will subscribe to a term deposit using machine learning.

Dataset

bank-full.csv with 45,211 records and 17 attributes.

Methodology

Data loading, preprocessing, EDA, feature engineering, Decision Tree training, evaluation, hyperparameter tuning, and model comparison.

EDA

Explored customer demographics, financial information, campaign attributes, and subscription trends.

Model

Decision Tree Classifier evaluated using Accuracy, Precision, Recall, F1-score, Confusion Matrix and ROC-AUC.

Conclusion

The project demonstrates a complete classification workflow suitable for business decision support.